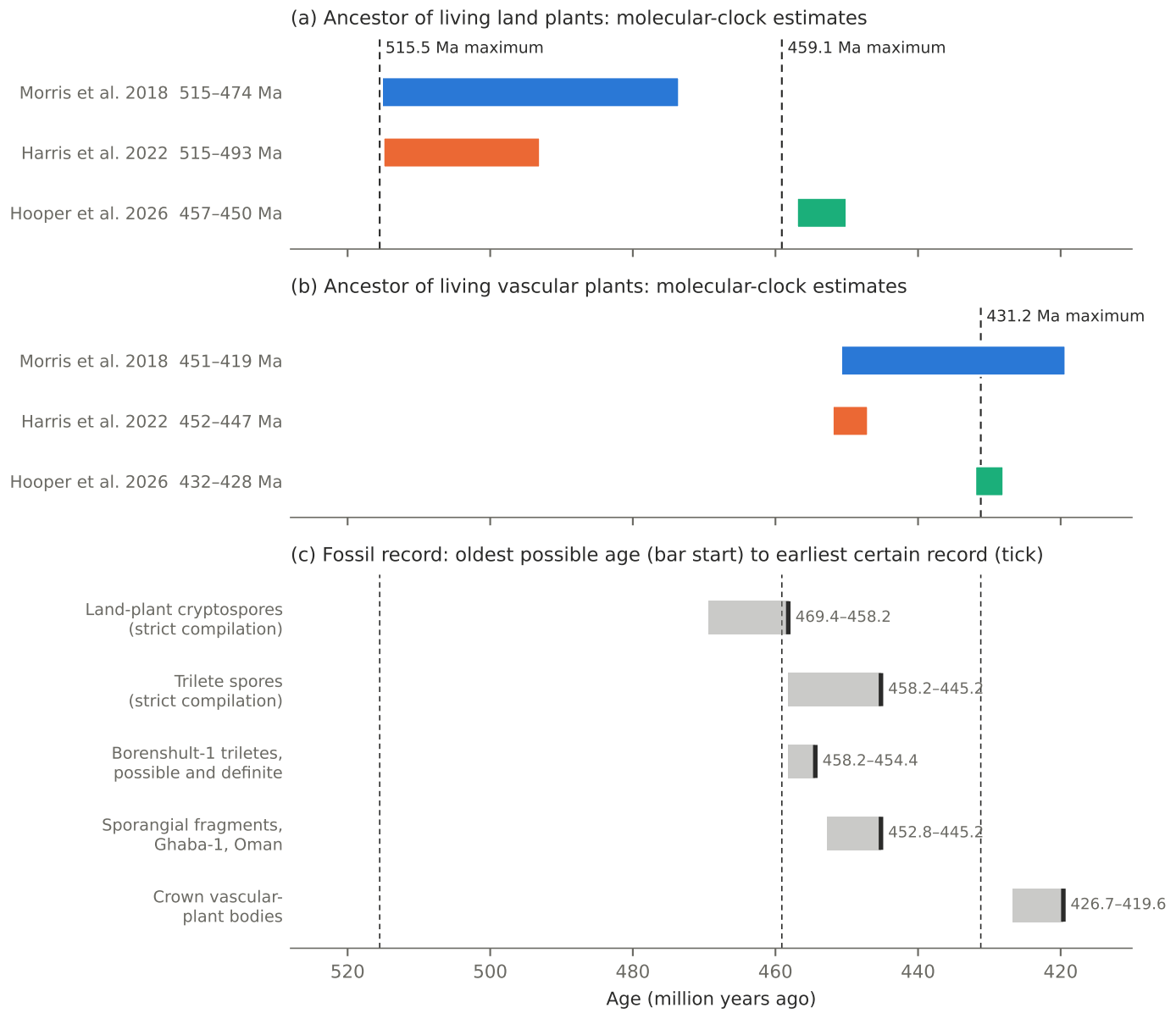
(a) Ancestor of living land plants: molecular-clock estimates
515.5 Ma maximum
459.1 Ma maximum
Morris et al. 2018 515–474 Ma
Harris et al. 2022 515–493 Ma
Hooper et al. 2026 457–450 Ma
(b) Ancestor of living vascular plants: molecular-clock estimates
431.2 Ma maximum
Morris et al. 2018 451–419 Ma
Harris et al. 2022 452–447 Ma
Hooper et al. 2026 432–428 Ma
(c) Fossil record: oldest possible age (bar start) to earliest certain record (tick)
Land-plant cryptospores (strict compilation)
469.4–458.2
Trilete spores (strict compilation)
458.2–445.2
Borenshult-1 triletes, possible and definite
458.2–454.4
Sporangial fragments, Ghaba-1, Oman
452.8–445.2
Crown vascular-plant bodies
426.7–419.6
520
500
480
460
440
420
Age (million years ago)